

SecureCloud (StegaVault)

Dual-Layer Zero-Knowledge Cryptographic & Steganographic Cloud Storage Platform

Domain:	Cybersecurity & Applied Cryptography
Core Ciphers:	AES-256-GCM + PBKDF2-HMAC-SHA256
Steganography:	1-Bit Spatial LSB PNG Embedding
Architecture:	Zero-Knowledge Client-Side WebCrypto Engine
Backend Tech:	NestJS · Prisma 7 · PostgreSQL (Supabase)
Date:	Academic Year 2026

1. Executive Summary & Problem Statement

1.1 The Fundamental Security Flaw in Traditional Cloud Storage

Mainstream cloud storage architectures (Google Drive, AWS S3, Dropbox, OneDrive) rely on **Server-Side Encryption (SSE)**. In this paradigm, while files are encrypted "at rest" on physical drives, the decryption keys are stored and managed within the cloud service provider's infrastructure.

This traditional model presents three severe threat vectors:

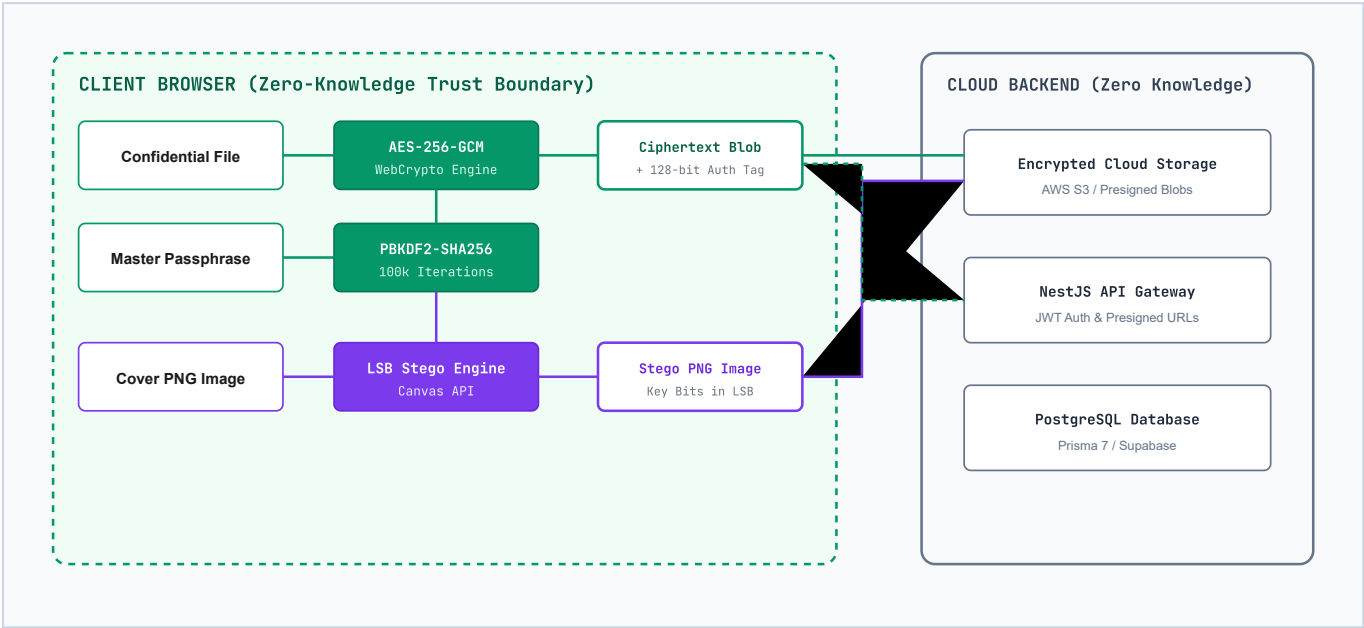
- **The Privileged Insider Threat:** System administrators, site reliability engineers (SREs), or database personnel with root infrastructure access can inspect plaintext user files.
- **Legal & Third-Party Compulsion:** Service providers can be legally compelled via court orders, subpoenas, or government national security letters to decrypt and yield customer data without the owner's knowledge.
- **Centralized Key Store Breaches:** If the cloud key management service (KMS) or authentication database is compromised, all tenant data across the entire platform is exposed simultaneously.

1.2 The SecureCloud Zero-Knowledge Solution

SecureCloud (StegaVault) enforces a mathematical **Zero-Knowledge Architecture**. All encryption and key generation operations occur strictly within the client device's browser sandbox using the hardware-accelerated W3C WebCrypto API.

The server infrastructure (NestJS API, PostgreSQL database, and object storage) only ever receives high-entropy ciphertext bytes and innocuous carrier images. Plaintext data and cryptographic master keys **never leave the client browser**.

2. System Architecture & High-Level Design



3. Core Algorithms & Mathematical Formulations

3.1 Key Derivation Function: PBKDF2-HMAC-SHA256

Human passwords possess low entropy. To derive a mathematically resilient 256-bit symmetric encryption key, SecureCloud uses **PBKDF2** (RFC 2898 / NIST SP 800-132).

```
DK = PBKDF2(PRF = HMAC-SHA256, Password = P, Salt = S, c = 100,000, dkLen = 256 bits)
```

- **Salt (\$\$):** 16 bytes of cryptographically secure pseudorandom numbers generated via `crypto.getRandomValues(new Uint8Array(16))`. Prevents rainbow table pre-computation.
- **Iteration Count (\$c = 100,000\$):** Imposes high computational cost on ASIC/GPU brute-force clusters while executing in under 50ms on modern client CPUs.

3.2 Authenticated Symmetric Encryption: AES-256-GCM

SecureCloud utilizes **AES in Galois/Counter Mode (GCM)** (NIST SP 800-38D). GCM is an *Authenticated Encryption with Associated Data (AEAD)* cipher that provides both **confidentiality** and **cryptographic integrity verification** in a single pipeline pass.

```
Ciphertext, AuthTag = AES-GCM-Encrypt(Key_256, IV_96, Plaintext_Data, AAD)
```

- **Cipher Standard:** 256-bit Advanced Encryption Standard (AES) Rijndael block cipher with 14 transformation rounds.
- **Initialization Vector (\$IV\$):** 96-bit unique random nonce per file to guarantee non-repeating counter keystreams.
- **128-Bit Authentication Tag (\$AuthTag\$):** Computed over the ciphertext using a Galois Field $\text{GF}(2^{128})$ polynomial multiplication (GMAC). If even a single bit of ciphertext is flipped or corrupted, decryption fails immediately with an integrity exception.

3.3 Spatial LSB (Least Significant Bit) Image Steganography

Instead of saving encryption keys in vulnerable plaintext files or key-stores, the key metadata (Header, Key, IV, and Auth Tag) is embedded into the **Least Significant Bits** of the RGB color channels of a PNG cover image.

```
Pixel_New = (Pixel_Original & 0xFE) | Secret_Bit
```

Imperceptibility Proof:

- An 8-bit color channel spans integers \$0\$ to \$255\$.

- Modifying the LSB shifts the color value by at most ± 1 , representing a maximal luminance change of: $\Delta L = \frac{1}{255} \approx 0.392\%$
- The human visual system (HVS) cannot distinguish luminance shifts below approximately $1.5\% - 2.0\%$ under standard display conditions, rendering the stego container visually identical to the original carrier image.

4. Comprehensive Threat Model & Security Comparison

Attack Vector / Threat	Standard Cloud Storage (Dropbox/S3)	SecureCloud (StegaVault)
Cloud Database Exfiltration	Total breach — Plaintext data exposed	Zero breach: Attacker obtains only random AES-256 ciphertext blobs
Malicious Cloud Administrator	Full visibility into files & keys	Zero visibility: Keys exist only in client browser memory
Ciphertext Bit-Flipping Attack	Silent corrupted decryption possible	Blocked: 128-bit GMAC Authentication Tag detects tampering
Cryptographic Key Discovery	Keys stored in central server vaults	Steganographic Hiding: Keys embedded inside PNG image LSBs
Brute-Force Passphrase Attack	Fast if unsalted or weak hashing	Hardware resistant: PBKDF2 100,000 rounds + 16-byte unique salt

5. Production Database Schema (Supabase PostgreSQL)

The backend persistence layer consists of 6 relational tables managed via Prisma 7:

Table Name	Primary Purpose	Key Security Attributes
users	User identity, Argon2 password hashes, MFA TOTP secrets	Role-based access (user/admin), brute-force lockout counters
oauth_accounts	Third-party OAuth links (Google, Discord)	Cascade deletion, unique provider-account indexing
files	Encrypted metadata & stego container object keys	Stores SHA-256 ciphertext checksums, IV, and Auth Tag
shares	Presigned secure file sharing links	Hashed token authentication, automatic time- based expiry
access_logs	Enterprise cryptographic audit logging	Real-time activity tracking, IP address, user agent audit trail
refresh_tokens	Session management & multi-device invalidation	SHA-256 hashed refresh tokens with 4-hour expiration window

6. Curated Academic & Video Learning Resources

The following verified educational video lectures explain the core computer science and cybersecurity principles applied in SecureCloud:

Core Concept	Recommended Video & Channel	Direct URL
AES-256 Encryption	<i>AES (Advanced Encryption Standard)</i> Spanning Tree	https://www.youtube.com/watch?v=C4ATDMIz5wc
Galois/Counter Mode (GCM)	<i>Galois/Counter Mode & AES</i> Computerphile	https://www.youtube.com/watch?v=04xNJsjtN6E
Image Steganography	<i>Secret Messages in Images (Steganography)</i> Computerphile	https://www.youtube.com/watch?v=TWEXCYQKyDc
LSB Hiding in Practice	<i>Steganography & LSB Hiding Tutorial</i> LiveOverflow	https://www.youtube.com/watch?v=5U_820c7m_E
Key Derivation & PBKDF2	<i>How NOT to Store Passwords!</i> Computerphile	https://www.youtube.com/watch?v=8ZtInCl5f0U
Zero-Knowledge & E2EE	<i>End-to-End Encryption in 100 Seconds</i> Fireship	https://www.youtube.com/watch?v=jkV1KEJGKRA